Modicon M580, M340, and X80 I/O Platforms

Standards and Certifications

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All pertinent state, regional, and local safety regulations must be observed when installing and using this product. For reasons of safety and to help ensure compliance with documented system data, only the manufacturer should perform repairs to components.

When devices are used for applications with technical safety requirements, the relevant instructions must be followed.

Failure to use Schneider Electric software or approved software with our hardware products may result in injury, harm, or improper operating results.

Failure to observe this information can result in injury or equipment damage.

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Safety Information

Important Information

NOTICE

Read these instructions carefully, and look at the equipment to become familiar with the device before trying to install, operate, service, or maintain it. The following special messages may appear throughout this documentation or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety label indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

⚠ DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

⚠ WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

⚠ CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result** in minor or moderate injury.

NOTICE

NOTICE is used to address practices not related to physical injury.

PLEASE NOTE

Electrical equipment should be installed, operated, serviced, and maintained only by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction and operation of electrical equipment and its installation, and has received safety training to recognize and avoid the hazards involved.

About the Book

At a Glance

Document Scope

This document presents the Standards and Certifications for the Modicon M580, M340, and X80 range.

Validity Note

This document is valid for Unity Pro 13.0 or later.

Product Related Information

⚠ WARNING

UNINTENDED EQUIPMENT OPERATION

The application of this product requires expertise in the design and programming of control systems. Only persons with such expertise are allowed to program, install, alter, and apply this product.

Follow all local and national safety codes and standards.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Introduction

The Modicon M580, M340, and X80 I/O platforms have been developed to comply with the principal national and international standards concerning electronic equipment for industrial automation systems.

- Requirements specific to programmable controllers: functional characteristics, immunity, resistance, safety, etc.: IEC/EN 61131-2 and IEC/EN/UL/CSA 61010-2-201, UL508
- Requirements specific to power utility automation systems: IEC/EN 61000-6-5, IEC/EN 61850-3
- Merchant navy requirements of the major international organizations: unified in IACS (International Association of Classification Societies)
- Compliance with European Directives for CE marking
- Low Voltage: 2014/35/EU
- Electromagnetic Compatibility: 2014/30/EU
- Ex areas:
  - For USA and Canada: Hazardous location class I, division 2, groups A,B,C, and D
  - For other countries: CE ATEX (2014/34/EU) or IECEx in defined atmosphere Zone 2 (gas) and/or Zone 22 (dust)
    Up-to-date information on which certifications have been obtained is available on our website.

M580, M340, and X80 I/O platforms are considered as open equipments and are designed for use in industrial environments, in pollution degree 2, over-voltage category II (IEC 60664-1), and in low-voltage installations, where the main power branch is protected on both wires by devices such as fuses or circuit breakers limiting the current to 15 A for North America and 16 A for the rest of the world.

Protective Treatment

The Modicon M580, M340, and X80 I/O platforms meet the requirements of TC treatment (Treatment for all Climates).

For installations in industrial production workshops or environments corresponding to TH treatment (Treatment for hot and humid environments), Modicon M580, M340, and X80 I/O must be embedded in enclosures with minimum IP 54 protection.

The Modicon M580, M340, and X80 I/O platforms offer protection to IP 20 level and protection against pins (enclosed equipment). In cases where a position is not occupied by a module, a BMXXEM010 protective cover must be installed.

They can therefore be installed without an enclosure in reserved-access areas which do not exceed pollution level 2 (control room with no dust-producing machine or activity). Pollution level 2 does not take account of more severe environmental conditions: air pollution by dust, smoke, corrosive or radioactive particles, vapors or salts, attack by fungi, insects, etc.

Treatment for more Severe Environments

If the platform has to be used in more severe environments or is required to start and operate in an extended temperature range, from –25 °C to +70 °C (–13 °F to +158 °F), select ruggedized equipments.

The ruggedized version of a standard equipment, that is hardened, or coated equipment, has a protective coating on its circuit boards.

This treatment increases the isolation capability of the circuit boards and their resistance to:

- Condensation
- Dusty atmosphere (conducting foreign particles)
- Chemical corrosion, in particular during use in sulphurous atmospheres (oil refinery, purification plant, etc.) or atmospheres containing halogens (chlorine, etc.)

This protection, combined with appropriate installation and maintenance, enables ruggedized equipments to be used in the following environments:

Harsh chemical environments

- **IEC/EN 60721-3-3 class 3C3:**
  - 14 days; 25 °C (77 °F) relative humidity 75%
  - Concentrations (mm^3/m^3): H_2S: 2100/SO_2: 1850/Cl_2: 100
- **ISA S71.04 classes G1 to G3:**
  - 14 days; 25 °C (77 °F) relative humidity 75%
  - Concentrations (mm^3/m^3): H_2S: 50/SO_2: 300/NO_2: 1250
- **IEC/EN 60068-2-52 salt mist, Kb test severity level 2:**
  - 3 x 24-hour cycles
  - 5% NaCl
  - 40 °C (104 °F) relative humidity 93%

Extreme climatic environments

- Temperature from –25 °C to +70 °C (–13 °F to +158 °F)
- Relative humidity levels:
  - up to 93% from –25 °C to +60 °C (–13 °F to +140 °F)
  - up to 95% from –25 °C to +55 °C (–13 °F to +131 °F)
- Formation of ice
- Altitudes from 0 to 5,000 m (0 to 16,404 ft)

Compliance with European Directives for CE marking.

Characteristics

Service Conditions and Recommendations relating to the Environment (Temperature/Humidity/Altitude)

Characteristics		Standard equipment	Ruggedized equipment
Temperature	Operation	0...+60 °C (+32...+140 °F)	-25...+70 °C (–13...+158 °F)
	Storage	- 40...+ 85 °C (–40...+185 °F)	- 40...+ 85 °C (–40...+185 °F)
Relative humidity (without condensation)	Cyclical humidity	+ 5...+95% up to 55 °C (+131 °F)	+ 5...+95% up to 55 °C (+131 °F)
	Continuous humidity	+ 5...+93% up to 55 °C (+131 °F)	+ 5...+93% up to 60 °C (+140 °F)
Altitude	Operation	• 0…2000 m (0...6562 ft): full specification: temperature and isolation • 2000...5000 m (6562...16404 ft): ❍ temperature derating: approx. 1°C/400 m (+34 °F/1312 ft) ❍ isolation: 150 Vdc/1000m (150 Vdc/3281 ft) For accurate temperature derating calculation, refer to IEC61131-2 Ed4.0 Annex A.	

Environment Tests

Overview

Standards and levels are given for the following tests:

- Immunity to low frequency interference *(see page 12)* (CE)
- Immunity to high frequency interference *(see page 13)* (CE)
- Electromagnetic emissions *(see page 15)* (CE)
- Immunity to climatic variations (Power ON) *(see page 16)*
- Withstand to climatic variations (Power OFF) *(see page 16)*
- Immunity to mechanical constraints (Power ON) *(see page 17)*
- Withstand to mechanical constraints (Power ON) *(see page 18)*
- Equipment and personnel safety *(see page 18)* (CE)
- Specific environment *(see page 19)*

NOTE: The tests with (CE) marking are required by European directives and based on IEC/EN 61131-2 standards.

Immunity to Low Frequency Interference (CE) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Voltage and frequency variations	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11	0.85...1.10 Un –0.94...1.04 Fn; 4 steps t = 30 min
	IACS E10; IEC 61000-4-11	0.80 Un...0.90 Fn; 1.20 Un...1.10 Fn; t = 1.5 s/5 s
Direct voltage variations	IEC/EN 61131-2; IEC 61000-4-29; IACS E10 (PLC not connected to charging battery)	0.85...1.2 Un + ripple: 5% peak; 2 steps t = 30 min
Third harmonic	IEC/EN 61131-2	H3 (10% Un), 0°/180°; 2 steps t = 5 min
Immunity to conducted low frequency (only IACS)	IACS E10	For AC: ● H2…H15 (10% Un), H15…H100 (10%...1% Un), H100…H200 (1% Un) For DC: ● H100…H200 (1% Un)
Voltage interruptions	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11; IEC 61000-4-29; IACS E10	Power supply immunity: ● 1 ms for DC PS1/10 ms, 85% Un for AC PS2 ● Check operating mode for longer interruptions up to 5s, 85% Un For IACS, 3 times 30 s in 5 mn, 85% Un
	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-11	For AC PS2: ● 20% Un, t0: ½ period ● 40% Un, cycle: 10/12 ● 70% Un, cycle: 25/30 ● 0% Un, cycle: 250/300
Voltage shut-down and start-up	IEC/EN 61131-2	● Un…0…Un; t = Un/60 s ● Umin…0…Umin; t = Umin/5 s ● Umin…0.9 Udl…Umin; t = Umin/60 s
Where: **PS1** applies to PLC supplied by battery **PS2** applies to PLC energized from AC or DC supplies **Un** nominal voltage **Fn** nominal frequency **Udl** detection level when powered		

Name of test	Standards	Levels
Magnetic field	IEC/EN 61131-2; IEC 61000-4-8 (for MV power stations: IEC 61000-6-5; IEC 61850-3)	Power frequency: 50/60 Hz, 100 A/m continuous...1000 A/m; t = 3 s; 3 axes
	IEC 61000-4-10	Oscillatory: 100 kHz...1 MHz, 100 A/m; t = 9 s; 3 axes
	IEC 61000-4-16 (for MV power stations: IEC 61000-6-5; IEC 61850-3)	For remote systems: ● 50/60 Hz and Vdc, 300 V, t = 1 s ● 50/60 Hz and Vdc, 30 V, t = 1 min ● 5 Hz…150 kHz, sweep 3…30 V
Where: **PS1** applies to PLC supplied by battery **PS2** applies to PLC energized from AC or DC supplies **Un** nominal voltage **Fn** nominal frequency **Udl** detection level when powered		

Immunity to High Frequency Interference (CE) Tests

NOTE: These tests are performed without an enclosure, with devices fixed on a metal grid and wired as per the recommendations in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Electrostatic discharges	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-2; IACS E10	6 kV contact; 8 kV air; 6 kV indirect contact
Radiated radio frequency electromagnetic field	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-3; IACS E10	10/15 V/m, 80MHz…1GHz; 3 V/m, 1.4…6 GHz Sinus amplitude modulated 80%,1 kHz + internal clock frequencies
Electrical fast transient bursts	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-4; IACS E10	For AC or DC main supplies: ● 2 kV in common mode/2 kV in wire mode For AC or DC auxiliary supplies, AC unshielded I/O: ● 2 kV in common mode For analog, DC unshielded I/O, communication and shielded lines: ● 1 kV in common mode

Name of test	Standards	Levels
Surge	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-5; IACS E10	For AC or DC main and auxiliary supplies, AC unshielded I/O: ● 2 kV in common mode/1 kV in differential mode For analog, DC unshielded I/O: ● 0.5 kV in common mode/0.5 kV in differential mode For communication and shielded lines: ● 1 kV in common mode
Conducted disturbances induced by radiated electromagnetic fields	IEC/EN 61131-2; IEC/EN 61000-6-2; IEC 61000-4-6; IACS E10	10 V; 0.15 MHz...80 MHz Sinus amplitude 80%, 1 kHz + spot frequencies
Damped oscillatory wave	IEC/EN 61131-2; IEC 61000-4-18; IACS E10	For AC or DC main supplies and AC auxiliary supplies, AC unshielded I/O: ● 2.5 kV in common mode/1 kV in differential mode For DC auxiliary supplies, analog, DC unshielded I/O: ● 1 kV in common mode/0.5 kV in differential mode For communication and shielded lines: ● 0.5 kV in common mode

Electromagnetic Emissions (CE) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Conducted emission	IEC/EN 61131-2; IEC/EN 61000-6-4; CISPR 11 and 22, Class A, Group 1 (FCC part 15 compliance)	150…500 kHz: quasi-peak 79 dB (µV/m); average 66 dB (µV/m) 500 kHz…30 MHz: quasi-peak 73 dB (µV/m); average 60 dB (µV/m)
	IACS E10	• AC or DC power (general power distribution zone): 10 kHz…150 kHz: quasi-peak 120...69 dB (µV/m); 150 kHz…0.5 MHz: quasi-peak 79 dB (µV/m) 0.5…30 MHz: quasi-peak 73 dB (µV/m) • AC or DC power (bridge and deck zone for evaluation): 10 kHz…150 kHz: quasi-peak 96…50 dB (µV/m) 150 kHz…0.35 MHz: quasi-peak 60…50 dB (µV/m) 0.35…30 MHz: quasi-peak 50 dB (µV/m)
Radiated emission	IEC/EN 61131-2; IEC/EN 61000-6-4; CISPR 11 and 22, Class A, Group 1 (FCC part 15 compliance)	30 MHz...230 MHz: quasi-peak 40 dB (µV/m) (at 10 m); 230 MHz...1 GHz: quasi-peak 47 dB (µV/m) (at 10 m); 1...3 GHz: quasi-peak 76 dB (µV/m) (at 3 m); 3...6 GHz: quasi-peak 80 dB (µV/m) (at 3 m);
	IACS E10	For general power distribution zone: • 0.15 MHz … 30 MHz: quasi-peak 80…50 dB (µV/m) (at 3 m) • 30...100 MHz: quasi-peak 60…54 dB (µV/m) (at 3 m) • 100 MHz...2 GHz: quasi-peak 54 dB (µV/m) (at 3 m) • 156…165 MHz: quasi-peak 24 dB (µV/m) (at 3 m)

Immunity to Climatic Variations (Power ON) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dry heat	IEC 60068-2-2 (Bb and Bd)	60 °C (+140 °F), t = 16 hrs (for ruggedized range: 70 °C (+158 °F), t = 16 hrs)
	IACS E10	60 °C (+140 °F), t = 16 hrs + 70 °C (+158 °F), t = 2 hrs (for ruggedized range: 70 °C (+158 °F), t = 18 hrs)
Cold	IEC 60068-2-1 (Ab and Ad); IACS E10	0 °C (32 °F), t = 16 hrs + power on at 0 °C (32 °F) (for ruggedized range: -25 °C (-13 °F), t = 16 hrs + power on at -25 °C (-13 °F))
Damp heat, steady state (continuous humidity)	IEC 60068-2-78 (Cab); IACS E10	55 °C (+131 °F), 93% relative humidity, t = 96 hrs (for ruggedized range: 60 °C (+140 °F))
Damp heat, cyclic (cyclical humidity)	IEC 60068-2-30 (Db); IACS E10	25…55 °C (77...131 °F), 93…95% relative humidity, 2 cycles t = 12 hrs +12 hrs
Change of temperature	IEC 60068-2-14 (Nb)	0 …60 °C (32...140 °F), 5 cycles t = 6 hrs + 6 hrs (for ruggedized range: -25…70 °C (77...158 °F))

Withstand to Climatic Variations (Power OFF) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dry heat	IEC/EN 61131-2; IEC 60068-2-2 (Bb and Bd); IEC/EN 60945	85 °C (185 °F), t = 96 hrs
Cold	IEC/EN 61131-2; IEC 60068-2-1 (Ab and Ad); IACS E10	- 40°C (-40 °F), t = 96 hrs
Damp heat, cyclic (cyclical humidity)	IEC/EN 61131-2; IEC 60068-2-30 (Db)	25…55 °C (77...131 °F), 93…95% relative humidity, 2 cycles t = 12 hrs + 12 hrs
Change of temperature (thermal shocks)	IEC/EN 61131-2; IEC 60068-2-14 (Na)	-40...85 °C (-40...185 °F), 5 cycles t = 3 hrs + 3 hrs

Immunity to Mechanical Constraints (Power ON) Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Sinusoidal vibrations	IEC/EN 61131-2; IEC 60068-2-6 (Fc)	Basic IEC/EN 61131-2: ● 5...150 Hz, ± 3.5 mm amplitude (5…8.4 Hz), 1 g (8.4…150 Hz) Specific profile: ● 5...150 Hz, ± 10.4 mm amplitude (5….8.4 Hz), 3 g (8.4….150 Hz) For basic and specific: endurance: ● 10 sweep cycles for each axis
	IEC 60870-2-2; IEC 60068-2-6 (Class Cm)	2...500 Hz, 7 mm amplitude (2…9 Hz), 2 g (9…200 Hz), 1.5 g (200…500 Hz) Endurance: 10 sweep cycles for each axis
	IACS E10	3...100 Hz, 1 mm amplitude (3…13.2 Hz), 0.7 g (13.2 …100 Hz) Endurance at each resonance frequency: 90 min for each axis, amplification coefficient < 10
	IEC 60068-2-6	Sismic analysis: 3...35 Hz, 22.5 mm amplitude (3…8.1 Hz), 6 g (8.1…35 Hz)
Shocks	IEC/EN 61131-2; IEC 60068-2-27 (Ea)	● 30 g, 11 ms; 3 shocks/direction/axis (1) ● 25 g, 6 ms; 100 bumps/direction/axis (bumps) (2)
Free fall during operation	IEC/EN 61131-2; IEC 60068-2-32 (Ed Method 1)	1 m (3.28 ft), 2 falls

(1) When using fast actuators (response time ≤ 5 ms) driven by relay outputs: 15 g, 11 ms; 3 shocks/direction/axis.
(2) When using fast actuators (response time ≤15 ms) driven by relay outputs: 6 ms; 100 bumps/direction/axis.

Withstand to Mechanical Constraints (Power OFF) Tests

Name of test	Standards	Levels
Random free fall with packaging	IEC/EN 61131-2; IEC 60068-2-32 (Method 1)	1 m (3.28 ft), 5 falls
Flat free fall	IEC/EN 61131-2; IEC 60068-2-32 (Ed Method 1)	10 cm (0.33 ft), 2 falls
Controlled free fall	IEC/EN 61131-2; IEC 60068-2-31 (Ec)	30° or 10 cm (0.33 ft), 2 falls
Plugging/Unplugging	IEC/EN 61131-2	For modules and connectors: Operations: 50 for permanent connections, 500 for non-permanent connections

Equipment and Personnel Safety Tests

NOTE: The devices must be installed, wired and maintained in accordance with the instructions provided in the *Grounding and Electromagnetic Compatibility of PLC Systems, Basic Principles and Measures, User Manual*.

Name of test	Standards	Levels
Dielectric strength and insulation resistance	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	Dielectric: 2 Un + 1000 V; t = 1 min Insulation: Un ≤ 50 V: 10 MΩ, 50 V ≤ Un ≤ 250 V: 100 MΩ
Continuity of earth	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	30 A, R ≤ 0,1Ω; t = 2 min
Leakage current	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	< 0.5 mA in normal condition < 3.5 mA in single fault condition
Protection offered by enclosures	IEC/EN 61131-2; IEC61010-2-201;	IP20 and protection against standardized pins
Impact withstand	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	Sphere of 500 g, fall from 1.30 m (4.27 ft) (energy 6.8 J minimum)
Overload	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	50 cycles, Un, 1.5 In; t = 1 s ON + 9 s OFF
Endurance	IEC/EN 61131-2; IEC 61010-2-201; UL; CSA	In, Un; 12 cycles: t=100 ms ON + 100 ms OFF, 988 cycles: t = 1 s ON + 1 s OFF, 5000 cycles: t = 1 s ON + 9 s OFF
Temperature rise	IEC/EN 61131-2; UL; CSA; ATEX; IECEx	Ambient temperature 60 °C (140 °F) (for ruggedized range: 70 °C (158 °F)) (1)
(1) Refer also to the chapter *Hardened Equipment*.		

Specific Environment Test

NOTE: Refer also to the chapter *Hardened Equipment*.

Name of test	Standards	Levels
Corrosion areas - gas, salt, dust	ISA S71.4	Flowing mixed gas; class Gx, 25 °C (77 °F), 75% relative humidity, t = 14 days
	IEC/EN 60721-3-3; IEC60068-2-60	Flowing mixed gas; class 3C3, 25 °C (77 °F), 75% relative humidity, t = 14 days
	IEC/EN 60721-3-3; IEC60068-2-60	Flowing mixed gas; class 3C4, 25 °C (77 °F), 75% relative humidity, t = 7 days
	IEC60068-2-52	Salt spray: test Kb, severity 2
	IEC/EN 60721-3-3; IEC60068-2-68	Dust and sand, Arizona dust, class 3S4, 20 cycles
	IEC/EN 60721-3-3; IEC60068-2-10	Mould growth, fungal spore, class 3B2, t = 28 days